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Institutional differences in awareness, attitude, utilization, and competence in artificial intelligence tools among chemistry lecturers and undergraduates in South-West Nigeria

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Abstract

This study investigated institutional differences in awareness, attitude, utilization, and competence regarding AI tools among chemistry lecturers and undergraduates in tertiary institutions in South-West Nigeria. A descriptive survey design was adopted, involving 300 lecturers and 600 undergraduates selected through a multi-stage sampling technique. Data were collected using validated structured questionnaires, Undergraduate Structured Questionnaire (USQ) and Lecturer Structured Questionnaire (LSQ), with reliability coefficients of 0.76 and 0.78 respectively. MANOVA results indicated no statistically significant multivariate effect of institutional type on lecturers' AI engagement, Pillai's Trace = 0.024, $F_{(8, 346)} = 0.53$, $p > .05$, partial $\eta^2 = 0.012$, nor among undergraduates, Pillai's Trace = 0.003, $F_{(8, 1734)} = 0.34$, $p > .05$, partial $\eta^2 = 0.002$. Univariate analyses showed negligible effect sizes across awareness, attitude, utilization, and competence. While awareness is gradually increasing, particularly among university lecturers and undergraduates in colleges of education, it has not yet translated into meaningful usage or confidence in applying AI tools for teaching and learning. In conclusion, the study calls for targeted capacity-building initiatives, curriculum reforms that embed AI literacy, and improved infrastructural support to foster effective utilization of AI tools in tertiary science education. Addressing these gaps is essential for preparing educators and undergraduates to navigate the evolving landscape of digital learning and to harness the full potential of AI in advancing chemistry education in Nigeria. The study recommends targeted training, curriculum integration, and infrastructural support to enhance AI adoption in science education.

Keywords Artificial Intelligence tools, AI adoption, Chemistry education, Tertiary institutions, Lecturers and undergraduates, Nigeria

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1 Introduction

In the evolving landscape of science education, Chemistry remains a cornerstone discipline that underpins technological advancement and national development. It provides a basis for understanding the composition, structure, properties, and behavior of matter [18]. The education of chemistry is vital in providing undergraduates with the essential knowledge and abilities to comprehend and contribute to scientific progress. Yet, conventional teaching techniques in chemistry frequently encounter difficulties in fully engaging undergraduates and fostering profound comprehension of intricate ideas. As such, there is a growing need to explore innovative approaches that can enhance chemistry education and improve student learning outcomes.

The Nigerian education system is structured into different levels, including primary, secondary, and tertiary education. Tertiary education in Nigeria consists of universities, polytechnics, and colleges of education. Universities are the highest level of academic institutions, offering undergraduates and postgraduate programs. Polytechnics focus on technical and vocational education, while colleges of education specialize in training teachers for primary and secondary schools. This research is on the tertiary institutions UNESCO [17].

The effectiveness of Chemistry instruction in Nigerian tertiary institutions is largely contingent upon the awareness, utilization, and competence of both lecturers and undergraduates in adopting modern pedagogical tools and resources. In the context of Nigerian tertiary institutions, awareness of instructional resources is a prerequisite for their effective utilization in teaching and learning [3]. Awareness encompasses recognition of available tools ranging from online databases and virtual laboratories to interactive simulations and understanding their pedagogical value. Without sufficient awareness, even well-resourced institutions struggle to modernize their curriculum and enhance student engagement.

Abubakar et al. [2] surveyed science lecturers across Colleges of Education in North-Central Nigeria and reported high levels of nominal awareness of virtual learning resources but markedly low practical engagement during classroom instruction. Mbonu-Adigwe and Obetta [10] found a similar pattern among tertiary science educators in Enugu State: while nearly all respondents knew about WhatsApp-based teaching platforms, fewer than half demonstrated the competence to integrate them effectively into lesson planning and delivery. Research focusing specifically on Chemistry teaching is more limited, yet parallels can be drawn from related STEM fields. Abdullahi and Oyelekan [1] examined senior school Chemistry teachers in Kwara Central and discovered that of forty-six identified online resources, teachers were fully aware of only one, leading to an acute underutilization of rich, freely available content such as PhET simulations and open-access journals. This underlines the need not only for raising awareness but also for reinforcing sustained use through professional development.

Understanding the current levels of awareness, attitude, utilization, and competence within Nigerian tertiary institutions is essential for shaping effective policies, professional development programs, and curriculum reforms. As the global shift toward digital and student-centered learning intensifies, the capacity of educators and learners to engage with innovative instructional methods becomes increasingly critical. Competence, which is defined as the ability to select, adapt, and deploy instructional resources to achieve specific learning outcomes is the third critical dimension. Competence grows

from a combination of training, hands-on experience, and institutional support [9]. Where training programs exist, educators report greater confidence in applying digital tools to complex concepts like molecular orbital theory or thermochemical calculations. However, uneven access to electricity, internet connectivity, and institutional incentives often stymies continuous skill development, perpetuating a cycle of low utilization despite growing awareness. This study seeks to explore these dimensions among Chemistry lecturers and undergraduates, aiming to bridge the gap between knowledge and practice in science education. This study has chosen artificial intelligence, as it is the recent innovation in education across the globe.

Artificial Intelligence (AI) is changing the way we teach and learn in tertiary institutions. In chemistry education, AI tools can help teachers explain difficult concepts better and allow undergraduates to learn in more personalized and interactive ways. These tools include things like smart tutoring systems, virtual labs, and software that helps with data analysis or writing reports. Research shows that many teachers and undergraduates still face challenges. For example, some chemistry teachers in Nigeria have low awareness of AI tools and rarely use them in class [16]. Other studies found that undergraduates who feel confident and have a positive attitude toward AI are more likely to use it [8]. Lecturers also struggle with using AI in research because of limited training and support [9].

Artificial Intelligence adoption in Nigeria is expanding rapidly across professional sectors. Recent reports indicate that approximately 69–72% of internet users in Nigeria have interacted with generative AI tools, representing tens of millions of users nationwide [7]. The highest levels of AI usage are recorded among ICT professionals, fintech workers, educators, students, healthcare practitioners, and digital entrepreneurs. In education, lecturers and students increasingly use AI for lesson preparation, research writing, and data analysis. Similarly, the finance sector uses AI for fraud detection and customer service automation, while healthcare professionals apply AI in diagnostic support systems. Despite this growing adoption, structured integration of AI into tertiary education remains limited, particularly in science disciplines such as chemistry, where awareness does not always translate into competence and utilization. This gap underscores the need for empirical studies assessing AI engagement among lecturers and undergraduates in Nigerian tertiary institutions [11].

AI tools offer great potential in education, many chemistry lecturers and undergraduates are still not making use of them. This may be due to limited knowledge about these tools, discomfort or hesitation in using them, or a lack of the necessary skills to apply them effectively. To better understand this situation, the study focuses on four key aspects: the level of awareness of AI tools, the attitudes toward their use, the frequency of actual usage, and the degree of competence in handling them. In Nigerian higher education, particularly in chemistry, most studies on AI adoption have focused on awareness and attitudes, but have paid little attention to how competence and actual use go hand in hand. Studies also tend to be limited in scope, focusing on single institutions or general education contexts, which restricts broader understanding [4]. Recent literature highlights the need for multivariate approaches that capture institutional differences and discipline-specific dynamics [4, 13]. The present study brings these constructs together within a single theoretical framework, allowing them to be examined not in isolation but as interdependent dimensions. By applying multivariate analysis, the study

moves beyond surface-level insights to provide a richer, more holistic understanding of AI adoption, particularly in the unique context of chemistry education in South-West Nigeria, where institutional differences and discipline-specific needs play a critical role.

1.1 Purpose of the study

This study investigated institutional differences in awareness, attitude, utilization, and competence in Artificial Intelligence tools among chemistry lecturers and undergraduates in South-West Nigeria. Specifically, this study investigated;

- The level of awareness, attitude, utilization and competence of lecturers across the Universities, Polytechnics and Colleges of Education in South-West Nigeria.
- The level of awareness, attitude, utilization and competence of undergraduates across the Universities, Polytechnics and Colleges of Education in South-West Nigeria.

1.2 Research questions

- What is the level of awareness, attitude, utilization and competence related to AI tools among chemistry lecturers across universities, polytechnics and colleges of education in South-West Nigeria?
- What is the level of awareness, attitude, utilization and competence related to AI tools among chemistry undergraduates across universities, polytechnics and colleges of education in South-West Nigeria?

1.3 Research hypotheses

- There is no significant difference between in the level of awareness, attitude, utilization and competence related to AI tools among chemistry lecturers across universities, polytechnics and colleges of education in South-West Nigeria.
- There is no significant difference between in the level of awareness, attitude, utilization and competence related to AI tools among chemistry undergraduates across universities, polytechnics and colleges of education in South-West Nigeria.

2 Theoretical framework

This study draws on the Technology Acceptance Model (TAM) and the Diffusion of Innovations Theory (DOI), both of which offer valuable insights into how new technologies are adopted and put to use in educational settings.

2.1 Technology acceptance model (TAM)

The technology acceptance model (TAM), introduced by Davis in 1989 [5] and refined by [12]; was designed to explain and predict how people come to accept and use information technologies. At its core, the model highlights two beliefs, Perceived Usefulness (PU) and Perceived Ease of Use (PEOU), as the main drivers of a person's attitude toward a technology. These attitudes then shape their intention to use it and, ultimately, their actual usage. Perceived usefulness reflects the belief that a technology will improve job performance, while perceived ease of use captures the sense that the technology requires little effort. Together, these perceptions influence how willing individuals are to adopt new tools and innovations. Decades of research continue to affirm TAM as one of the

most powerful and widely applied frameworks for understanding technology adoption in both educational and organizational contexts.

In this study, awareness of AI tools is seen as the first step being the initial exposure to new technologies. Attitude captures how users evaluate and feel about adopting AI, while utilization reflects the actual integration of these tools into teaching and learning. Competence, in turn, represents the skills and confidence needed for sustained, effective use. According to TAM, lecturers and undergraduates who view AI tools as both useful and easy to use are more likely to develop positive attitudes, engage with them more often, and build stronger competence over time. This makes TAM a solid theoretical foundation for exploring differences in AI awareness, attitudes, utilization, and competence among chemistry lecturers and undergraduates.

Recent studies have reinforced the relevance of the Technology Acceptance Model (TAM) in today's educational technology research, showing its strong ability to predict both users' intentions and their actual engagement with digital tools such as artificial intelligence, virtual learning systems, and other emerging platforms. As a result, TAM serves as a valuable framework for examining the factors that shape AI adoption in tertiary chemistry education.

2.2 Diffusion of innovations theory (DOI)

The diffusion of innovations theory, first introduced by Rogers [14] and later refined by Dearing and Cox [6], explains how new ideas and technologies spread within a social system over time through communication and interaction. The theory outlines five stages of adoption: knowledge, persuasion, decision, implementation, and confirmation. In practice, this means individuals first learn about an innovation, form opinions about it, decide whether to adopt it, put it into use, and then either continue or abandon its use based on their experience.

The Diffusion of Innovations Theory also categorizes adopters into five groups, innovators, early adopters, early majority, late majority, and laggards, each reflecting different rates at which people embrace new technologies. In educational contexts, this diffusion process is shaped by factors such as institutional support, access to resources, professional development, and peer influence. For AI tools to be successfully integrated into chemistry education, it is not enough for lecturers and undergraduates to simply be aware of them; progress through the stages of adoption and the development of lasting competence are equally critical.

In this study, the constructs under investigation align closely with the stages of the Diffusion of Innovations Theory. Awareness reflects the knowledge stage, attitude corresponds to persuasion, utilization represents implementation, and competence aligns with confirmation, where users demonstrate sustained and effective use of AI technologies. Taken together, the theory offers a valuable framework for understanding how AI tools are introduced, disseminated, and adopted across universities, polytechnics, and colleges of education.

2.3 Integration of TAM and DOI in the present study

Bringing TAM and DOI together offers a well-rounded explanation of AI adoption among chemistry lecturers and undergraduates. TAM sheds light on the psychological factors that shape individual acceptance and use of AI technologies, while DOI

highlights the broader social and institutional processes through which these innovations spread and become embedded in educational practice. Combined, the two theories provide a complementary framework for analyzing awareness, attitude, utilization, and competence in relation to AI tools. They also support the exploration of institutional differences in adoption across tertiary institutions in South-West Nigeria, where variations in infrastructure, training opportunities, and organizational support may significantly influence both the diffusion and acceptance of emerging technologies.

3 Methodology

3.1 Research design

This study employed a descriptive survey design to investigate awareness, attitude, utilization, and competence regarding artificial intelligence (AI) among chemistry lecturers and undergraduates in tertiary institutions across South-West Nigeria. A quantitative approach was adopted, using structured questionnaires to collect data.

3.2 Population and sampling

The population comprised 596 full-time chemistry lecturers and 5,786 chemistry undergraduates across six federal and state institutions (Colleges of Education, Polytechnics, and Universities) located in Ekiti, Ondo, Oyo, Osun, Ogun, and Lagos States.

A multi-stage sampling technique was used to ensure representativeness. Institutions were purposively selected to cover all six states. Departments (Chemistry and Chemistry Education) were chosen through stratified random sampling to balance lecturers and undergraduates. Individual respondents were selected using systematic sampling.

The sample size for the study was determined using the Taro Yamane formula, which is specifically designed for finite populations. According to this formula, the required sample size (n) is calculated as:

$$n = \mathrm{N}/(1 + \mathrm{N}(\mathrm{e}^2))$$

Where N represents the total population size, and e denotes the level of precision or margin of error, which in this case was set at 0.05 (5%). This formula provides a simplified and statistically sound method of ensuring that the chosen sample is representative of the population while maintaining a reasonable level of accuracy. At a 5% margin of error, the minimum required sample sizes were 239 lecturers and 374 undergraduates. To strengthen representativeness and account for non-response, the sample was expanded to 300 lecturers and 600 undergraduates, exceeding the minimum requirement and enhancing statistical power.

3.3 Instruments

Two instruments were developed for the study: the Lecturer Structured Questionnaire (LSQ) and the Undergraduate Structured Questionnaire (USQ). Each instrument consisted of forty items, carefully divided into four sections. The first section measured awareness of AI tools, the second examined attitudes toward AI tools, the third assessed utilization of AI tools, and the fourth evaluated competence in using AI tools, with each section containing ten items. Responses were collected using a four-point Likert scale, ranging from Strongly Agree (4) to Strongly Disagree (1). To ensure consistency in interpretation, a benchmark mean of 2.50 was adopted as the decision rule, representing the

midpoint of the scale. Based on this criterion, scores were classified into three levels of engagement. Scores between 1.00 and 1.99 indicated low levels of engagement or understanding, scores between 2.00 and 2.49 reflected moderate familiarity and usage, while scores from 2.50 to 4.00 signified high levels of engagement and competence in the use of AI tools. This design provided a structured and quantifiable means of capturing both lecturers' and undergraduates' perspectives, ensuring that the data collected could be meaningfully analyzed to reveal patterns of awareness, attitudes, utilization, and competence in relation to AI tools.

3.4 Validity and reliability

Validity was established through expert review by specialists in measurement, chemistry education, and educational technology. Exploratory Factor Analysis (EFA) confirmed construct validity, with a Kaiser-Meyer-Olkin (KMO) value of 0.86 and a significant Bartlett's Test of Sphericity ($\chi^2 = 3245.61$, $p < .05$). Principal Component Analysis with Varimax rotation extracted four clear factors, awareness, attitude, utilization, and competence, with loadings ranging from 0.54 to 0.82.

Reliability was assessed using Cronbach's alpha, yielding high internal consistency shown in Table 1.

3.5 Data collection and analysis

Ethical clearance was obtained, and informed consent was secured from all participants. Questionnaires were distributed in both paper and electronic formats. For analysis, mean and standard deviation were used to answer research questions, given the descriptive nature of Likert-scale data. To test hypotheses, Multivariate Analysis of Variance (MANOVA) was employed, as it allows simultaneous examination of multiple dependent variables (awareness, attitude, utilization, competence) across institutional types. This approach reduces Type I error and provides richer multivariate insights.

The decision rules guiding the analysis were clearly defined to ensure consistency and objectivity in interpreting the results. A mean score of 2.50 or greater was considered acceptable, while a mean score below 2.50 was regarded as rejected. In testing the hypotheses, statistical significance was determined using the p-value. Specifically, any hypothesis with a p-value less than 0.05 was rejected, indicating a statistically significant difference, whereas hypotheses with a p-value equal to or greater than 0.05 were upheld, signifying no significant difference. These rules provided a structured framework for evaluating both descriptive and inferential statistics in the study. All analyses were conducted using IBM SPSS Statistics version 25.

3.6 Result

Table 2 describes the demographic composition of the study population.

Table 1 Reliability coefficients of awareness, attitude, utilization and competence for both undergraduates and lecturers

Variable	Undergraduates (α)	Lecturers (α)
Awareness	0.76	0.78
Attitude	0.76	0.81
Utilization	0.76	0.76
Competence	0.77	0.79
Overall	0.76	0.78

Table 2 Demographic characteristics of study population

Variable	Category	Lecturers ($n=300$)	Undergraduates ($n=600$)	Total (%)
Institution Type	Universities	100	200	33.3
	Polytechnics	100	200	33.3
	Colleges of Education	100	200	33.3
Gender	Male	178	312	54.4
	Female	122	288	45.6
Teaching Experience	0–5 years	74	–	24.7
	6–10 years	92	–	30.7
	Above 10 years	134	–	44.6
Level of Study	100–200 level	–	215	35.8
	300–400 level	–	385	64.2

Table 3 Lecturers' awareness, attitude, utilization and competence

School type	Awareness	Attitude	Utilization	Competence	Expected mean
Colleges of education	1.69	1.77	1.08	1.08	2.50
Polytechnics	1.58	1.75	1.25	1.02	
Universities	1.83	1.90	1.12	1.17	

Table 3 above show notable differences across institutions in South-West Nigeria. University lecturers demonstrated the highest levels of awareness and competence, as well as the most positive attitudes toward AI tools. Colleges of Education followed closely, while polytechnics consistently ranked lowest in awareness and competence, though they reported slightly higher utilization than the other groups.

Despite these variations, all mean scores across awareness, attitude, utilization, and competence remain well below the expected benchmark of 2.50. This indicates that while university lecturers are relatively more aware and optimistic, and polytechnic lecturers show marginally greater use, overall engagement with AI tools in chemistry instruction is still limited.

Table 4 compares differences based on institution type (university, polytechnic, college). Regarding chemistry lecturers' level of awareness, where p value is 0.460. Institution type doesn't influence awareness level. Attitude towards AI tools is also not significant with p value of 0.125. For utilization, p value is 0.575, there is no significant difference and also for competence where p values is 0.214, there is no significant difference also. Hence the null hypothesis is upheld.

The data satisfied the assumption of multivariate normality based on inspection of skewness and kurtosis values within acceptable limits. Homogeneity of covariance matrices was assessed using Box's M test, which was non-significant, indicating equality of covariance matrices across groups. Additionally, Levene's test of equality of error variances for each dependent variable was not significant, supporting the assumption of homogeneity of variances.

The multivariate test showed that institution type did not significantly influence the combined dependent variables. Using Pillai's Trace, there was no statistically significant multivariate effect of institution type on lecturers' AI awareness, attitude, utilization, and competence, *Pillai's Trace* = 0.024, $F_{(8, 346)} = 0.53$, $p > .05$, partial $\eta^2 = 0.012$. Similarly, Wilks' Lambda indicated a non-significant multivariate effect, $\Lambda = 0.976$, $F_{(8, 344)} = 0.52$, $p > .05$, partial $\eta^2 = 0.002$. These results suggest that institution type does not jointly affect lecturers' engagement with AI tools.

Table 4 MANOVA showing lecturers' awareness, attitude, utilization and competence of AI tools across South-West tertiary institutions.

Source	Dependent variable	Type III Sum of Squares	df	Mean square	F	Sig.
Corrected Model	Level of Awareness	.850[a]	2	0.425	0.781	0.460
	Attitude towards AI tools	.435[b]	2	0.218	2.106	0.125
	Level of Utilization	.214[c]	2	0.107	0.555	0.575
	Level of Competence	.400[d]	2	0.200	1.554	0.214
Intercept	Level of Awareness	156.292	1	156.292	287.097	0.000
	Attitude towards AI tools	176.203	1	176.203	1704.835	0.000
	Level of Utilization	71.407	1	71.407	369.562	0.000
	Level of Competence	63.205	1	63.205	491.489	0.000
School	Level of Awareness	0.850	2	0.425	0.781	0.460
	Attitude towards AI tools	0.435	2	0.218	2.106	0.125
	Level of Utilization	0.214	2	0.107	0.555	0.575
	Level of Competence	0.400	2	0.200	1.554	0.214
Error	Level of Awareness	95.268	175	0.544		
	Attitude towards AI tools	18.087	175	0.103		
	Level of Utilization	33.814	175	0.193		
	Level of Competence	22.505	175	0.129		
Total	Level of Awareness	675.000	178			
	Attitude towards AI tools	649.000	178			
	Level of Utilization	261.000	178			
	Level of Competence	259.000	178			
Corrected Total	Level of Awareness	96.118	177			
	Attitude towards AI tools	18.522	177			
	Level of Utilization	34.028	177			
	Level of Competence	22.904	177			

[a]R Squared = 0.009 (Adjusted R Squared = – 0.002)

[b]R Squared = 0.024 (Adjusted R Squared = 0.012)

[c]R Squared = 0.006 (Adjusted R Squared = – 0.005)

[d]R Squared = 0.017 (Adjusted R Squared = 0.006)

Follow-up univariate ANOVA results further confirmed the absence of statistically significant differences across institution types for all dependent variables. Lecturers' level of awareness did not differ significantly across institutions, $F_{(2, 175)} = 0.781$, $p > .05$, with a very small effect size (partial $\eta^2 = 0.009$). Attitude towards AI tools also showed no significant difference, $F_{(2, 175)} = 2.106$, $p > .05$, with a small effect size (partial $\eta^2 = 0.024$). Similarly, utilization of AI tools was not significantly different across institution types, $F_{(2, 175)} = 0.555$, $p > .05$ (partial $\eta^2 = 0.006$). Lecturers' competence in using AI tools also did not vary significantly across institutions, $F_{(2, 175)} = 1.554$, $p > .05$ (partial $\eta^2 = 0.017$).

The effect sizes for all dependent variables were small, indicating that institution type explains only a minimal proportion of variance in lecturers' awareness, attitude, utilization, and competence regarding AI tools. This suggests that lecturers across universities, polytechnics, and colleges of education demonstrate relatively similar levels of engagement with AI technologies.

Beyond the absence of statistical significance, the findings imply that exposure to AI tools among chemistry lecturers may be shaped more by individual factors such as personal interest, professional development opportunities, and access to digital infrastructure rather than institutional classification. The consistency across institution types may also reflect the increasing diffusion of AI technologies across tertiary education settings in South-West Nigeria, where lecturers share comparable professional networks, training platforms, and digital environments.

Table 5 Undergraduates' awareness, attitude, utilization and competence

School type	Awareness	Attitude	Utilization	Competence	Expected mean
Colleges of education	1.88	1.34	1.15	1.55	2.50
Polytechnics	1.84	1. 33	1.18	1.48	
Universities	1.81	1.32	1.14	1.45	

Table 6 MANOVA showing undergraduates' awareness, attitude, utilization and competence of AI tools across South-West tertiary institutions

Source	Dependent variable	Type III sum of squares	df	Mean square	F	Sig.
Corrected Model	Level of Awareness	.344[a]	2	0.172	0.425	0.654
	Attitude towards AI	.035[b]	2	0.018	0.080	0.923
	Level of Utilization	.090[c]	2	0.045	0.186	0.830
	Level of Competence	.765[d]	2	0.382	1.079	0.340
Intercept	Level of Awareness	1336.600	1	1336.600	3298.939	0.000
	Attitude towards AI	697.046	1	697.046	3165.492	0.000
	Level of Utilization	524.536	1	524.536	2167.388	0.000
	Level of Competence	875.631	1	875.631	2471.165	0.000
School	Level of Awareness	0.344	2	0.172	0.425	0.654
	Attitude towards AI	0.035	2	0.018	0.080	0.923
	Level of Utilization	0.090	2	0.045	0.186	0.830
	Level of Competence	0.765	2	0.382	1.079	0.340
Error	Level of Awareness	352.490	870	0.405		
	Attitude towards AI	191.575	870	0.220		
	Level of Utilization	210.551	870	0.242		
	Level of Competence	308.275	870	0.354		
Total	Level of Awareness	3256.000	873			
	Attitude towards AI	1725.000	873			
	Level of Utilization	1363.000	873			
	Level of Competence	2177.000	873			
Corrected Total	Level of Awareness	352.834	872			
	Attitude towards AI	191.611	872			
	Level of Utilization	210.641	872			
	Level of Competence	309.040	872			

[a]R Squared = 0.001 (Adjusted R Squared = – 0.001

[b]R Squared = 0.000 (Adjusted R Squared = – 0.002)

[c]R Squared = 0.000 (Adjusted R Squared = – 0.002)

[d]R Squared = 0.002 (Adjusted R Squared = 0.000)

Table 5 above shows that across South-West Nigerian tertiary institutions, chemistry undergraduates show only modest familiarity with AI tools. Colleges of Education report the highest levels of awareness, attitude, and competence, with polytechnics close behind and universities slightly lower. However, all scores remain below the expected benchmark, indicating that awareness has not matured into meaningful engagement.

Utilization is minimal across all institutions, with polytechnic students showing marginally greater use than their peers. Competence levels follow a similar pattern, with Colleges of Education leading but still far below satisfactory standards.

From Table 6, a one-way Multivariate Analysis of Variance (MANOVA) was conducted to determine whether undergraduates' awareness, attitude, utilization, and competence in AI tools differed across institution types (university, polytechnic, and college of education) in South-West tertiary institutions. Prior to analysis, assumptions were evaluated. The data met the assumption of multivariate normality, as skewness and kurtosis values fell within acceptable thresholds. Homogeneity of covariance matrices was

assessed using Box's M test, which was non-significant, indicating that covariance matrices were equal across groups. Furthermore, Levene's tests for equality of error variances for each dependent variable were non-significant, confirming homogeneity of variance.

The multivariate test indicated that institution type did not significantly influence the combined dependent variables. Using Pillai's Trace, there was no statistically significant multivariate effect of institution type on undergraduates' awareness, attitude, utilization, and competence in AI tools, *Pillai's Trace* = 0.003, $F_{(8, 1734)} = 0.34$, $p > .05$. Similarly, Wilks' Lambda showed no significant multivariate effect, $\Lambda = 0.997$, $F_{(8, 1732)} = 0.34$, $p > .05$. These findings indicate that institution type does not jointly affect undergraduates' engagement with AI technologies.

Follow-up univariate ANOVA results revealed no statistically significant differences across institution types for all dependent variables. Undergraduates' level of awareness did not significantly differ across institutions, $F_{(2, 870)} = 0.425$, $p > .05$, with a negligible effect size (partial $\eta^2 = 0.001$). Attitude towards AI tools also showed no significant difference, $F_{(2, 870)} = 0.080$, $p > .05$, with an extremely small effect size (partial $\eta^2 \approx 0.000$). Likewise, utilization of AI tools did not differ significantly across institutions, $F_{(2, 870)} = 0.186$, $p > .05$ (partial $\eta^2 \approx 0.000$). Competence in AI tools also showed no significant difference across institution types, $F_{(2, 870)} = 1.079$, $p > .05$, with a very small effect size (partial $\eta^2 = 0.002$).

The effect sizes across all dependent variables were negligible, indicating that institution type explains virtually none of the variance in undergraduates' awareness, attitude, utilization, and competence regarding AI tools. This suggests that undergraduates across universities, polytechnics, and colleges of education exhibit comparable engagement levels with AI technologies.

Beyond statistical non-significance, these findings imply that exposure to AI tools among undergraduates may be influenced more by personal access to digital devices, internet connectivity, peer learning, and informal technology use rather than institutional classification. The uniformity across institution types may reflect widespread accessibility of AI platforms, particularly mobile-based tools, which allow students to interact with AI technologies regardless of their institutional setting.

3.7 Discussion of findings

This section discusses the major findings of the study, provides plausible explanations, and compares the results with existing literature.

University lecturers demonstrated the highest awareness and optimism toward AI tools, followed by lecturers in colleges of education and polytechnics. Despite this, their utilization and competence remained low. University lecturers may have greater exposure to academic discourse and professional development opportunities, which fosters awareness. However, without structured training and institutional policies, awareness does not necessarily lead to effective use. Thomas et al. [15] found that qualifications and motivation, rather than institutional type, influence AI usage. This aligns with the present findings, where awareness exists but practical competence lags. Idika et al. [9] emphasized institutional barriers such as access and training, which may explain why awareness has not translated into utilization.

Undergraduates in colleges of education reported slightly higher awareness and competence compared to their peers in universities and polytechnics. Nonetheless,

enthusiasm, usage, and confidence remained low across all groups. Colleges of education may emphasize pedagogical training, which could encourage students to explore educational technologies. However, limited infrastructure and lack of integration into coursework hinder sustained engagement. Thomas et al. [15] argued that individual motivation is more critical than institutional type, which resonates with the finding that undergraduates' engagement is uniformly low. Idika et al. [9] suggested that institutional barriers, such as inadequate access to AI tools, can restrict both lecturers and students, implying that institutional type might indirectly influence outcomes.

Institutional type did not significantly affect lecturers' or undergraduates' perceptions and engagement with AI tools. R-squared values were extremely low, showing that school type explains virtually none of the variation in awareness, attitude, utilization, or competence. The null hypothesis was upheld. This suggests that systemic challenges, such as lack of training, infrastructure, and institutional policies, are common across universities, polytechnics, and colleges of education, overshadowing any differences in institutional type. Thomas et al. [15] confirmed that institutional affiliation is not a determining factor, emphasizing instead the role of qualifications and motivation. Conversely, Idika et al. [9] observed disparities across departments and genders, indicating that while institutional type may not matter broadly, specific contexts within institutions can shape AI engagement.

3.8 Educational implications

The findings imply that while awareness of AI tools is gradually emerging among lecturers and undergraduates, actual usage and competence remain low, pointing to the need for systemic interventions. Institutions must invest in training, curriculum integration, and infrastructure to move beyond superficial awareness and foster meaningful engagement. Since institutional type does not significantly influence outcomes, emphasis should be placed on individual motivation, equitable access, and supportive policies across all tertiary institutions. Without these measures, AI adoption in education will remain limited and fail to translate into improved teaching and learning practices.

3.9 Limitation of the study

This study was limited to selected tertiary institutions in South-West Nigeria, which may restrict the generalizability of the findings to other regions. Additionally, the use of self-reported questionnaires may introduce response bias. To minimize these limitations, a multi-stage sampling technique was employed to enhance representativeness, and the research instruments were validated and tested for reliability to improve data accuracy.

4 Conclusion

AI awareness is growing, particularly among university lecturers and undergraduates in colleges of education, but this has not yet translated into meaningful usage or competence. Institutional type does not significantly influence AI engagement among lecturers or undergraduates. Individual factors like motivation, academic qualification, and departmental culture may play a more critical role in shaping AI adoption. There is a pressing need to move beyond awareness and foster practical engagement and skill development in AI tools across all tertiary institutions.

4.1 Recommendations

The following recommendations were raised from the findings. It is recommended that tertiary institutions organize regular training workshops to enhance lecturers' and students' competence in using artificial intelligence tools. Curriculum developers should integrate AI-based instructional strategies into chemistry education to promote effective utilization. Government and institutional management should provide adequate technological infrastructure to support AI adoption. Additionally, collaborative research initiatives should be encouraged to explore discipline-specific applications of artificial intelligence in chemistry teaching and learning.

Author contributions

Peter Olubunmi Olugbuyi: Conceptualization, methodology design, data analysis, and manuscript drafting.Mulkah Adebisi Ahmed: Statistical analysis, interpretation of findings, and final manuscript editing.Adekunle Solomon Olorundare: Instrument development, data collection coordination, and literature review.All authors read and approved the final manuscript.

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Data availability

The datasets generated and analyzed during the current study are available from the corresponding author on reasonable request.

Declarations

Ethics approval and consent to participate

This study received ethical clearance from the Science Education Research Ethics Committee in accordance with the Declaration of Institutional Research Ethics of Nigeria (IREN). All procedures involving human participants were conducted in accordance with institutional and national ethical standards. The protocol, with the approval number (EKSU/EDU/SEDREC/159) was approved by Science Education Research Ethics Committee, Ekiti State University, Ado Ekiti in accordance with the principles of the Declaration of Institutional Research Ethics of Nigeria (IREN). This research involved human participants only. No animals were used in any part of the study. All human-related procedures complied with ethical standards set by the relevant institutional and national research committees. All participants provided written or electronic consent to participate in the study. Participation was voluntary, and respondents were assured of confidentiality and anonymity.

Consent for publication

The authors affirm that participants consented to the publication of anonymized data and findings derived from the study.

Competing interests

The authors declare no competing interests.

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